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NIR photometry of crowded fields and identification and characterization of variable stars - A study of Carina dSph

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NIR photometry of crowded fields and identification and characterization of variable stars - A study of Carina dSph

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Abstract

We present new and unpublished near-infred data for the Carina dwarf spheroidal galaxy, collected using the 6.5 m Baade telescope's FourStar Camera at Las Campanas (Chile). We present the dataset and the observing strategy, which is different from the classical optical data. We describe the pre-reduction operations performed on the dataset using the IRAF package, to avoid the problems linked to the NIR photometry. We also describe the followed reduction process performed thanks to the DAOPHOT/DAOMATCH/DAOMASTER/ALLFRAME suites, and the calibration chosen to produce the CMD. We found that our CMD is $\simeq$ 1.5 mag deeper than the calibration catalog and is comparable with the most recent literature works (Karczmarek et al. 2021). To the isochrone analysis and to the variable stars identification will be given particular attention. The first is consistent with the previous literature works, while for the latter a cross-match with the catalog from Coppola et al. 2015 found 73 variable stars. We present new near infrared mean magnitudes for 67 of them and we verify the RR Lyrae PLZ relation, comparing the results with theoretical relations and obtaining a distance modulus of $DM_0 = 20.08 \pm 0.04$ mag.

1 Introduction

Dwarf galaxies are the most common type of galaxies in the universe and the simplest galactic systems known (Mateo 1998). They contain up a few million of stars and usually orbit larger galaxies. This could be verified observing the Local Group (LG), i.e. the portion of our universe with a diameter $\simeq$ 3Mpc which includes the Milky Way (MW) and its satellites, the Andromeda galaxy (M31) and its satellites and a consistent number of "neighbour galaxies" which are not bounded to the MW or to M31.
There are few characteristics and properties that made the dwarf galaxies an interesting target in the past decades:

- they subtend large areas on the sky and contain a high density of stars, many of them faint which had been resolved thanks to the advent of wide-field digital astronomy (McConnachie 2012)
- as the simplest galactic system known, they could enrich our vision of the galactic hierarchy and maybe give new insights on the Hubble classification
- they could host a broad variety of variable stars
- They tend to be low-luminosity and low-metallicity making them a prototype of galaxies in the early Universe
- They could give new insight on galaxy formation (and disruption) and on the role of dark matter.

For the reasons listed above the dwarf galaxies are both an excellent stellar and observational astrophysical laboratory, and a low-redshift ($z \simeq 0$) cosmological laboratory. The latter property is particularly interesting. Dwarf galaxies, with respect to globular clusters, presents multiple star formation events, separated in time and in metallicities involved, and show the presence of a large fraction of dark matter. This is why they've been also used to test the modern ΛCDM cosmological models (for example Walker et al. 2009). However, simulations performed using the same models show that the number of expected dwarf satellites of the MW should be far higher than the number of dwarfs observed, and this is called "the missing satellites problem" (Bullock 2010).
For these reasons keep observing dwarf galaxies is extremely important. In this context the NIR photometry could play an important role for many reasons: (i) it is the next reasonable step in the observational history of these low-luminosity astrophysical objects; (ii) has various observational and theoretical advantages as the weak reddening dependence and the increase of the period-luminosity relations slope for RRL; (iii) only a few dwarf galaxies have precise local standards in the optical and in NIR bands.

1.1 The case of study: Carina dSph

Spheroidal dwarf galaxies (dSph) are a subset of the dwarf galaxies previously introduced. The prototype of this class of objects is the Sculptor dSph (Shapley 1938), which in the discovery paper has been addressed as a "a symmetrical Magellanic Cloud" or "a nearby spheroidal galaxy, highly resolved, and of abnormally low surface brightness". The object of interest of this report is the Carina dSph, a satellite of the MW discovered in 1977 (Cannon et al. 1977). It is relatively close to our galaxy and its central density is modest and so it could be highly resolved, Figure 1. Physical parameters from McConnachie 2012 are listed in Table 1.

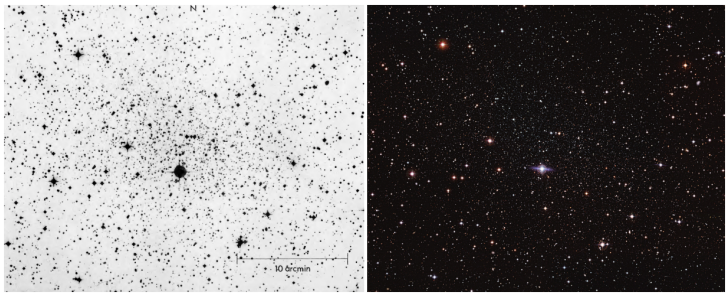


Figure 1: *Left*: Image of the Carina dSph taken with the UK 1.2-m Schmidt Telescope in 1977, year of the discovery (Cannon et al. 1977). *Right*: Image of the Carina dSph made using observations from the Wide Field Imager on the MPG/ESO 2.2-metre telescope at La Silla, and from the Victor M. Blanco 4-metre telescope at the Cerro Tololo Inter-American Observatory. (https://www.eso.org/public/usa/images/potw1126a/?lang)

From the photometric point of view, Carina dSph has been studied in deep by: Dall'Ora et al. 2003, Monelli et al. 2003 and Stetson et al. 2011, providing optical CMD and studying the variable stars. The results showed the presence of a broad variety of variable stars , from Cepheids to RR Lyrae, and even Anomalous Cepheids (Dall'Ora et al. 2003). Regarding the optical CMD, it clearly shows multiple star formation events, separated in time and stellar mass involved (Monelli et al. 2003). Following studies, which involved stellar isochrones from the BaSTI database [1] (Pietrinferni et al. 2004) showed that is possible to identify two separate stellar population: (i) a young populations with age $t = 5 \pm 2$ Gyr, and (ii) an old one with $t = 12 \pm 1$ Gyr, as shown by Bono et al. 2010 and Stetson et al. 2011. Moreover, the alpha elements enhanced models with [Fe/H]=- 1.84 dex, [M/H]=- 1.79 dex and [α/H]= + 0.4 dex showed a better agreement, especially regarding the RGB branch.

In the last decade, Carina dSph has been studied also using NIR photometry, and thanks to the Peter Stetson's private database [2] is possible to resume some of the observations. More recently, the Araucaria Project (Karczmarek et al. 2021) presented a deep near infrared map. The current state of the art is shown in Figure 3. Using the Stetson data is also possible to verify that Carina dSph is characterized by two main star formation episodes, plotting an optical-NIR CMD ($V, V-J$) against the isochrones and ZAHB locus from the BaSTI Database. We chose to not modify the magnitudes in the CMD, i.e. to not plot the reddening-corrected magnitudes but to add the correct extinction coefficients to the theoretical curves. The color excess term (and so the A_i coefficients) has been computed thanks to

[1] http://basti.oa-teramo.inaf.it/index.html
[2] https://www.canfar.net/storage/list/STETSON/Standards

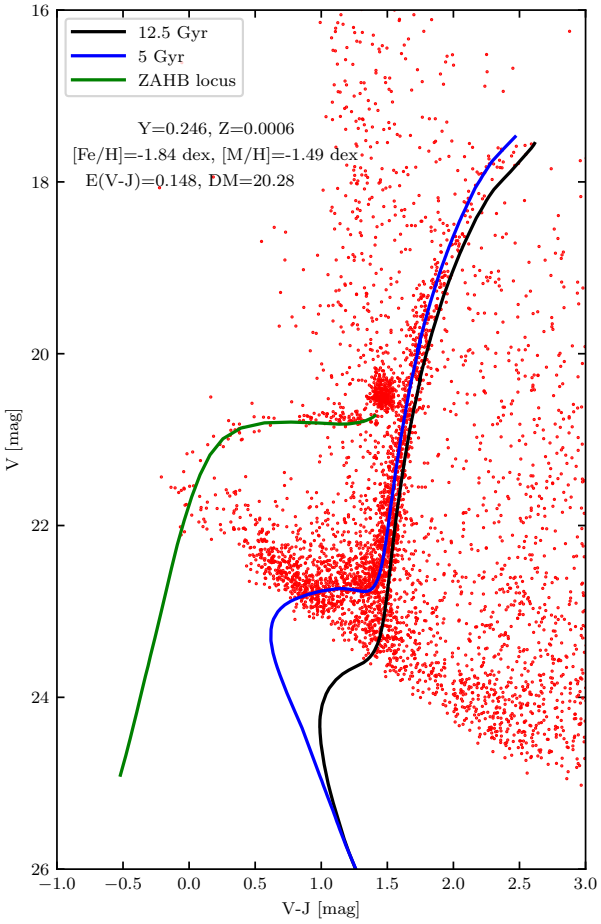


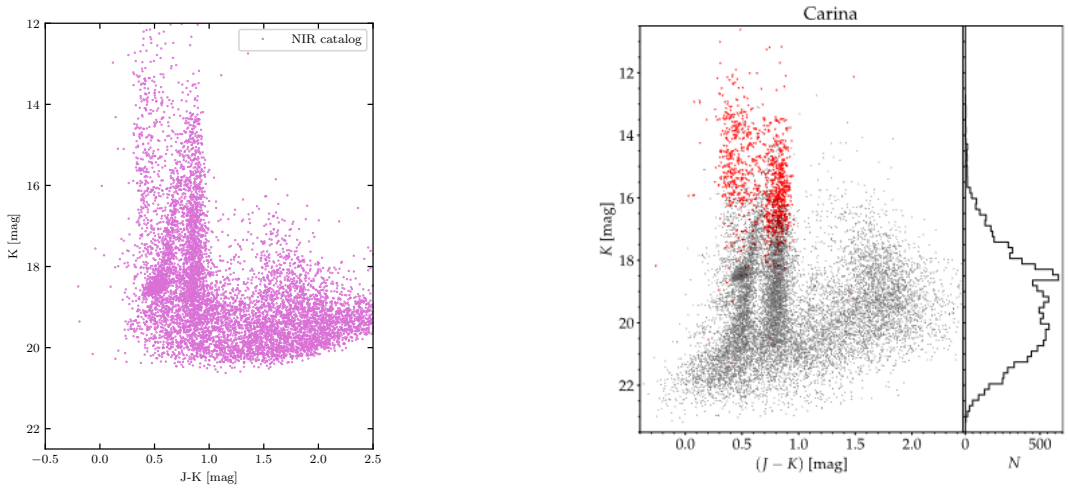
Figure 2: Optical-NIR CMD of Carina dSPh form the Peter Stetson's Private Database is shown with red dots, while theoretical isochrones from the BaSTI database are shown in black and magenta. In green is shown the ZAHB locus, always from the BaSTI and with the same metallicities of the isochrones curves.

Parameter	Symbol	Value	Reference
Right Ascension	RA	$06^h41^m36.7^s$	Cannon et al. 1977
Declination	Dec.	$-50^h57^m58^s$	Cannon et al. 1977
Galactic Longitude	l	260.1	Pietrzyński et al. 2009
Galactic Latitude	b	-22.2	Pietrzyński et al. 2009
Color Excess	$E(B-V)$	0.06	NASA/IPAC IRSA
Distance Modulus	$(m-M)_0$	20.01 ± 0.02	Coppola et al. 2015
Galactocentric Velocity	v_{MW}	$7\ \mathrm{km \cdot s^{-1}}$	M. G. Walker et al. 2009
Apparent Visual Magnitude	V	11.0 ± 0.5	Irwin et al. 1995
Absolute Visual Magnitude	M_V	-9.1 ± 0.5	Irwin et al. 1995
Half-Light Radius	r_h (pc)	250 ± 39	Irwin et al. 1995
Ellipticity	ϵ	0.33 ± 0.05	Irwin et al. 1995
Central Surface Brightness	μ_V (mag/")	25.5 ± 0.5	Irwin et al. 1995
Mass	M_G ($10^6\ M_\odot$)	0.38	Walker et al. 2009
Velocity Dispersion	σ ($\mathrm{km \cdot s^{-1}}$)	6.6 ± 1.2	M. G. Walker et al. 2009

Table 1: Physical parameters of the Carina dSph, following the same reasoning from McConnachie 2012

the IRSA's Dust Reddening and Extinction Calculator [3]. The results are shown in Figure 2, and are consistent with the previous literature works as discussed above. It is worth to underline that the value named DM is the value (in [mag]) which has been added to the theoretical curves to fit the experimental data. As explained above this is not the true distance modulus but the apparent distance modulus DM_V, and this notation will be adopted also in Section 5. The value DM_V=20.28 mag is consistent with previous literature works, if we consider for example that the true distance modulus is $\mathrm{DM}_0 = 20.11$ mag (Dall'Ora et al. 2003), and $A_V \simeq 0.2$ mag from NASA IRSA.

Every measurement could improve our knowledge of Carina dSph, and this report follows this reasoning. In section 2 we present the new unpublished NIR data of Carina dSph and we describe the instrument used to collect them and the observational strategy. In section 3 we present the advantages and disadvantages of NIR photometry and explain the strategy used to avoid the latter ones and to improve the quality of the raw data. In section 4 we discuss the reduction and calibration strategy, and in section 5 we present the results of the reduction analysis.



(a) (b)

Figure 3: The current state of the art for Carina dSph NIR photometry. (a): K, $J-K$ CMD of Carina using the data from the Peter Stetson's private database database; (b): K, $J-K$ CMD of Carina dSph from Karczmarek et al. 2021. The background stars are shown in red

2 Observations

The near-infrared dataset we present and analyze, were collected the night of 2013/09/22, with the FourStar Infrared Camera installed at the 6.5-m Baade Telescope built by the Carnegie Institution of Washington at its Las Campanas Observatory in Chile on behalf of the Magellan Project [4], see Figure 4.

The telescope has an altazimuth mount and the FourStar Camera is placed at the Naysmith East focus. FourStar Camera is composed by four CCD arrays 2048×2048, with a field of view of 10.8'×10.8' and a pixel scale of 0.159"/pixel. It is possible to identify three major components (Monson et al. 2014):

1. *The optics:* consist of a seven lenses system L1-L7. L1 is the dewar window, L2 focuses the telescope entrance and L7 is the field flattener. The lenses L3-L6 focus the beam onto the detectors with a ratio f/3.6. All the system is held at 200 K thanks to a radiation shield.

2. *The detectors plane:* There are four Teledyne HAWAII2-RG 2048x2048 detectors in a close-pack configuration, for a total of 16 Mpixels. The gaps

[3]https://irsa.ipac.caltech.edu/applications/DUST/
[4]https://obs.carnegiescience.edu/Magellan

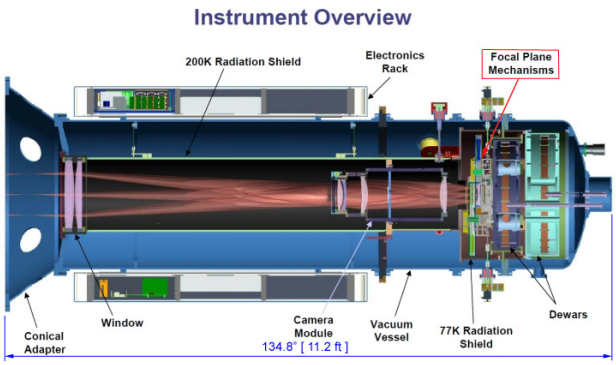


Figure 4: Overview of the FourStar Infrared Camera where is possible to observe the optical system, the camera focal plane and the temperature and pressure control system.

Array ID	Serial Number	Well Depth (e^{-1})	Gain (e^{-1}/ADU)	Read Out Noise (e^{-1})
1	SN192	$1.55 \cdot 10^6$	2.65	25.5
2	SN204	$1.43 \cdot 10^6$	2.59	22.1
3	SN209	$1.43 \cdot 10^6$	2.51	20.5
4	SN216	$1.36 \cdot 10^6$	2.49	18.9

Table 2: Physical properties of the CCD arrays of the FourStar Infrared Camera installed at the 6.5-m Baade Telescope at Las Campanas (Chile). The data in the Table refer to operation mode labelled "FullWell" which has been selected to collect the photometric data presented in Section 2, and have been taken from the FourStar official manual (Monson et al. 2014).

between the arrays are about 19 arc-seconds. There are two working mode and the data were taken using the one labelled "FullWell". Physical parameters are listed in Table 2. All the arrays are held at 77 K thanks to another radiation shield.

3. *The pressure and temperature control system:* Has the function of keeping the optics and the detectors at the right temperature. It is composed by two dewars: the shroud dewar and the detector dewar, both containing liquid nitrogen (LN2). Moreover, on the dewars are placed sixteen diodes which monitor the internal temperatures of the instrument, via two Lakeshore model 340 temperature controllers operated by a "meta-controller" system running on the instrument's control computer.

The dataset consist of 534 jittered images of Carina dSph, 240 in the H filter and 294 in the J filter. The J filter is centered at 1.25 μm, and the H filter is centered at 1.61 μm, both with a width of $\simeq 0.4\,\mu m$, and so within the wavelenght range defined as near infrared NIR (0.7-3 μm). The jittering, i.e. the random shift

of the telescope for a given field, was performed to avoid sky-level variations and saturation, which affects the NIR domain (see Section 3). Referring to Figure 5, the camera needed four different pointings to cover all the Carina dSph region. In every pointing were performed 5 jitters and the measures were repeated three times in a time interval comparable with the exposure time (each measure is called *Loop).* In this way we can cover all the Carina dSph region using 20 images for every filter, for every array and for every loop. The exposure time of every image was 11.644 s. Some fields in the J were observed more than three times, having in this way a few hours coverage. It is worth to underline that the dithering process often used by large ESO telescopes (see for example Karczmarek et al. 2021) was not performed to obtain the images.
The dataset also includes a series of 60 Flat Field images taken at the twilight (to achieve a uniform illumination for ideally every pixel) in the J band (20 images with $t_{\rm exp} = 5.822\ s$, and 40 images with $t_{\rm exp} = 11.644\ s$), a series of 20 Flat Field images taken at the twilight in the H band with $t_{\rm exp} = 5.822\ s$ and a series of 80 Dark Images (i.e. with the shutter closed) with $t_{\rm exp} = 1.456\ s$.

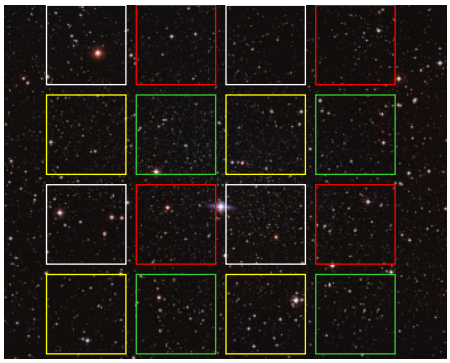


Figure 5: Representation of the observational strategy followed to collect the data presented in Section 2. The choice of colors is the following: red for the array #1, green for the array #2, yellow for the array #3 and white for the array #4. Every box refers to a series of 5 images taken centering the telescope in a point and then shifting it by $\simeq 10^2$ pixels. In this way is possible to map all the Carina dSph region using short exposure images to avoid saturation. It is worth to underline that the boxes did not represent the true pointing of the telescope, but are used to simplify the presentation of the data.

3 Pre-reduction

The term pre-reduction refers to every procedure aimed at removing instrumental effects on raw scientific images. Before exposing our pre-reduction strategy, it is worth to briefly discuss some aspects of NIR observations. First, it must be noticed that IR arrays are different from the optical ones, usually referred only as CCD. IR arrays are made with semi-conductors with a smaller energy gap between conduction and valence bands, with respect to optical CCDs. For this reason, as it is shown in Figure 4, the arrays must be held at low temperatures, to avoid the interference of thermal signal. Modern arrays, as the HAWAII2-RG of the FourStar Camera (Loose et al. 2003), are now CMOS sensors.
Knowing this difference, is possible to list some of the advantages of NIR observations with respect to optical observations:

- In IR arrays and CMOS each pixel is effectively isolated from his neighbors, avoiding problems as blooming, charge trail and bleeding
- is less affected by the reddening, i.e. the absorption of the interstellar medium (Cardelli et al. 1989)

However, there are also disadvantages, as:

- The radiation background from the sky and observing environment is much higher, and is not constant both spatially and temporally. Moreover, due to the detector characteristics, often non-linearity regimes and saturation are reached sooner for IR arrays
- IR arrays are more expensive and difficult to build
- A significant part of IR radiation is blocked by the Earth's atmosphere (see for example the telluric lines)
- Observations are strongly seeing limited.

A first read may lead to the conclusion that NIR observations are highly disadvantaged with respect to optical ones. However, it must be reminded that reddening is a problem that affects astronomy since its origins, and when limited has produced incredible results, as the measurement of the MW bulge star's high proper motion, which in turn lead to the discovery of the supermassive black hole at the center of our galaxy (Ghez et al. 2008). It is worth to underline that a kind of measure would have been impossible with optical observations. On the other hand, it is mandatory to avoid all the difficulties, and this justifies both the jittering process explained in Section 2 and the short exposure times.

Usually, all pre-reduction procedures requires the use of Flat Fields and Dark images to get rid of instrumental and background problems simultaneously. A basic pre-reduction strategy could be: $I = \frac{I_{\rm raw} - D}{FF}$, where FF represents a flat field, D a dark image and I the pre-reduced image. However NIR images require a more complex strategy and the one we followed can be divided in various steps, each performed thanks to the IRAF software (Tody 1986) [5]:

1. *Linearity correction:* All real CCDs and arrays have a linear response in a given range in ADU. The FourStar arrays in "FullWell" mode are linear until $\simeq 2 \cdot 10^4$ ADU. Monson et al. 2014, found that the deviation from linearity is a single term of order 2.5: $t = A_0 \cdot S + A_1 \cdot S^{2.5} + A_2$, where t represents the exposure time and S the raw counts. True counts are given by t/A_0. Using the values pubblished by the FourStar collaboration [6] and the IRAF command **imexpr** is possible to linearize each of the 534 images and all the dark images.

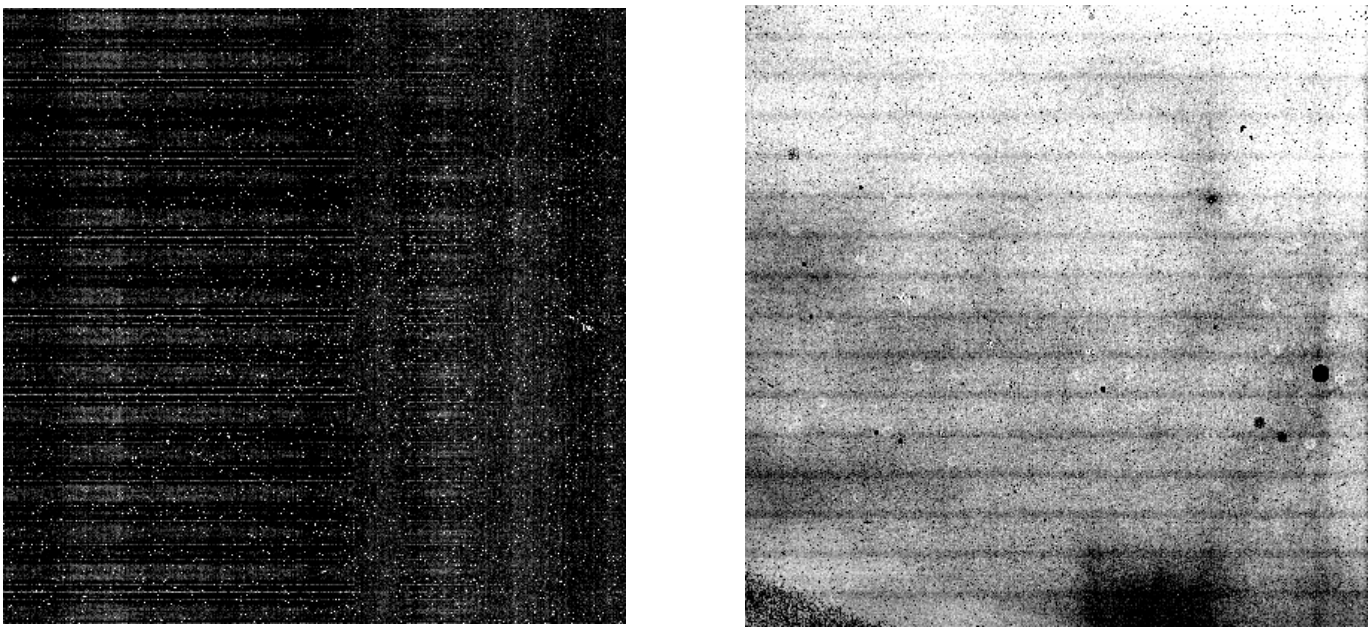
2. *Dark preparation:* As we saw background is variable both spatially and temporally, so a reasonable step is the one in which all the Dark Images are averaged together and then subtracted (solving the temporal dependence). To do that, we used first the IRAF command **imstat** to eliminate the corrupted dark images, and then the IRAF command **imcombine**. The averaged Dark Image could be seen in Figure 6a.

3. *Flat Fields Preparation:* To avoid all the problems connected to pixel and field non-uniformity, vignetting and fixed pattern noise we must use Flat Field images. For the same reasons of the Step 2, is more convenient to work with an averaged Flat Field for each of the band (H and J). To obtain them we must: (i) use the IRAF command **imarith** to multiply the averaged Dark Image for a number n such that $t_{\rm dark} \cdot n = t_{\rm FF,i}$, where $t_{\rm dark} = 1.456\,s$ and $t_{\rm FF,i}$ is the exposure time of the considered flat field image (ii) use the IRAF command **imarith** to subtract the nth multiplied Averaged Dark Image to the Flat Field Images with the same exposure time (iii) use the IRAF command **imcombine** and average all the corrected Flat Fields (iv) use the IRAF command **imcopy** to trim the averaged Flat Fields to avoid edge imperfections (v) use the IRAF command **imnorn** to normalize them. The averaged, trimmed and normalized flat field for the H band could be observed in Figure 6b. It is worth to underline that the subtraction between Flat Fields and Dark images get rid of the bias and of background temporal dependence.

[5] https://iraf-community.github.io/

[6] http://instrumentation.obs.carnegiescience.edu/FourStar/calibration.html

4. *SofiRedu3 Analysis:* The effects we need to remove yet are the spatial dependence of the background and the cosmetic effects, i.e. the effects due to corrupted pixels. Both the effects could be removed using the IRAF pipeline **sofiredu3**, written by Massimo Dall'Ora. For a given image, the pipeline could iteratively generate a background image using all the other images taken with the same filter, and also subtract it together with the averaged dark and the bad pixel mask, normalizing every step with the right Flat Field Image. The bad pixel mask is a .fits image which contains all the information about the corrupted pixels in each array and has been published by the FourStar Collaboration together with the linearity correction coefficients.

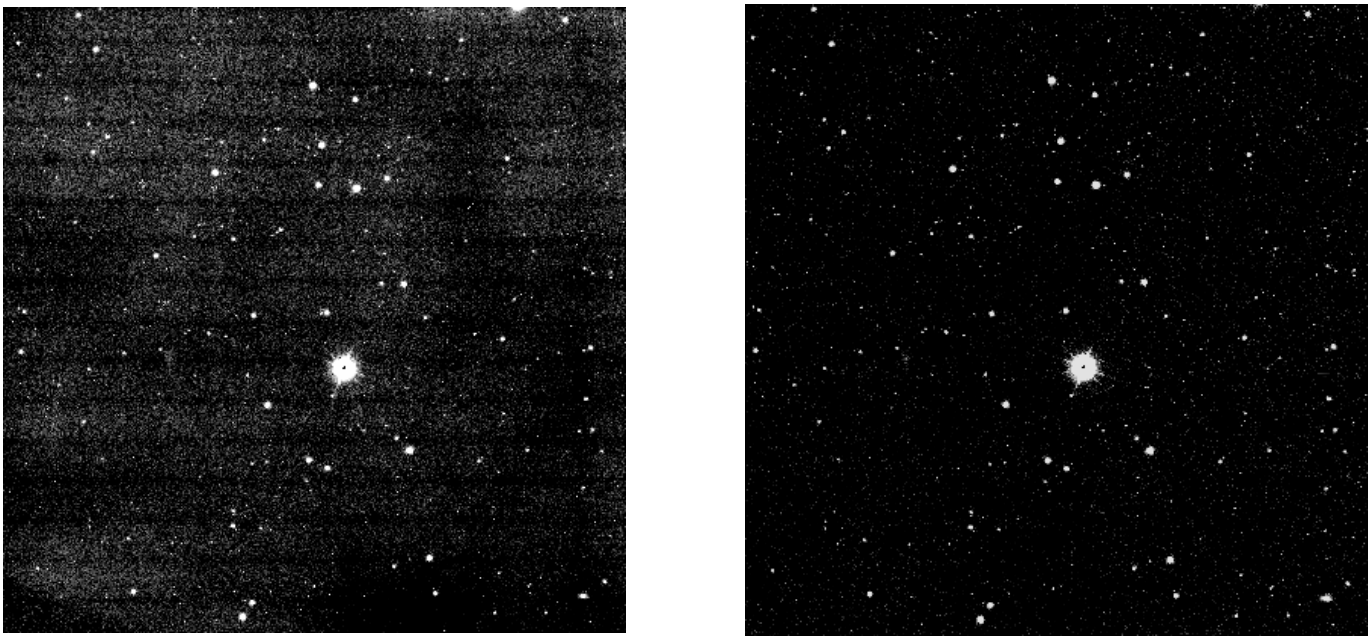


(a) (b)

Figure 6: *Left*: The averaged Dark Image for the FourStar Camera's third array. The process to obtain the image is discussed in Section 3. *Right*: The averaged and normalized Flat Field Image for the FourStar Camera's first array. The process to obtain the image is discussed in Section 3.

Figure 7 shows the results of the pre-reduction process on an image taken with the J filter. Lastly, each of the pre-reduced image was astrometrized thanks to the codes produced by the Astrometry.net Collaboration [7] (Lang et al. 2010).

[7] https://astrometry.net/



(a) (b)

Figure 7: An example of the action of **sofiredu3**, before (a) and after (b), on an image of a part of Carina dSph taken in the *J* band and the first array of the FourStar Camera. The image on the left is not a raw image because it has been linearized.

4 Reduction and calibration

For the reasons exposed in the previous Section, it is reasonable to expect that also NIR photometry reduction requires a series of techniques which differs from the optical case. In particular due to the high backgroung noise, PSF photometry is one of the best way to produce valuable scientific data. This, essentially, is aimed at find the PSF with which the images have been taken, which is a function of both the instrumental properties and the seeing conditions. The PSF represents quantitatively how much a point-like source is spreaded in the detector plane, and could be used to discriminate stars from background objects. A quick way to verify this could be using the IRAF command **imexam** and select both a bright star and a possible galaxy. In this case the FWHM of the two objects will differ significantly.
Moreover, if the images have been acquired using a jittering process is important to bear this in mind during the reduction, or significant errors could be committed. As in the case of pre-reduction (Section 3), our strategy could be divided in various steps:

1. *PSF Photometry:* carried out thanks to the standalone programs DAOPHOT and ALLSTAR (Stetson 1987). DAOPHOT is composed by various tasks and works using the chosen image and an option file as input. The more important tasks in our analysis are: (i) **FIND** which finds all the peaks brighter than 3σ of the local background (ii) **PHOT** which performs an aperture photometry with chosen aperture radii, (iii) **PICK** which selects well-suited stars for the calculus of the PSF from the ones found with the previous tasks and (iv) **PSF** which performs the PSF photometry fit for the picked stars with a chosen point spread function profile (which has been left has a free parameter) and produces a residuals file (a .psf file). Moreover, DAOPHOT could model spatially varying PSFs, iteratively adding an empirical correction (from the residuals) to the analytic function. Then ALLSTAR performs the PSF fit (with the same properties) for all the stars of the given image, re-determining the centroid of the stars and re-calculating the sky background after that all the stars have been provisionally removed by the fit. The result of this process are two files (a file .psf and a file .als) and an image with the residuals. Using a script we performed iteratively the process described above for each of the 534 pre-reduced images, re-calculating at each step the PSF to find the best agreement (for example the script at each step removes the corrupted picked PSF stars, which are labelled by DAOPHOT at the end of the **PSF** task).

2. *Coadd the Images:* Once the PSF photometry is performed for an image, the next step is to cross-correlate the measurements with the ones of the other 533 images. So, finding the geometric relation between each image is essential, as it is a reference image. The chosen reference image is a co-adding of all the 534 images, and has been produced using the MONTAGE2 program (Stetson 1993). In this way we obtained a co-added image without second order effects due to super-imposition of some images, as it is shown in Figure 8. However, in this way the image lost the astrometric correction explained in Section 3, so we generated a second co-added image using SWarp (Bertin et al. 2002), which in our case did not produce a good quality image as the one produced with MONTAGE2, but kept the astrometric information.

3. *Geometric Transformations Search:* We first used **FIND** and then ALLSTAR on the co-added image, to have a reference list of stars, which is fundamental to find the geometric transformation. In fact, we used DAOMATCH (Stetson 1993), which is a program that works on files generated with ALLSTAR. The way we used the program is quite manual, in fact having the MONTAGE2 image as a reference, and the SWarp image as an

help, we selected the jittered images for every *Loop* and we found the corresponding pixel region in the MONTAGE2 image. With this information is possible to run DAOMATCH, otherwise the program would have not converged. This is caused by the fact that DAOMATCH works on a master frame image (.als) and on a series of second images, for which it founds the geometric transformations in the X and Y space (image coordinates) in term of rotations and translations. The process will converge and produce optimal result if and only if we are working on images with similar sizes. At the end we found 16 matches, which have been cut in a single file (this is possible because every match refers to the same .als file). At this point we refined the matches using DAOMASTER (Stetson 1993), which is a program that could essentially produce a star catalog. The program has three fundamental parameters: (i) **minimum number of frames**, i.e. the number of frames in which a star should be detected to be kept in your output catalog, (ii) **minimum fraction of frames**, i.e. the fraction of input frames in which a source should appear, and (iii) **enough frames**, i.e. the number of frames that in which source must be detected to stay in your catalog. We run DAOMASTER twice to ensure that every star would be contained at least in one frame .

4. *First ALLFRAME Analysis:* At this point was possible to run ALLFRAME (Stetson 1994), which is a program that makes simultaneous use of the geometric and photometric information from ALL images of a given field to derive a self-consistent set of positions and magnitudes for all the detected star-like objects in that area of sky.

5. *Residuals Analysis :* ALLFRAME gives back a series of residuals images. We co-added them using MONTAGE2 as in the second step. In this way is possible to study the faintest sources which have not been detected by ALLFRAME. To do that we used the task **FIND** and **PSF**, and then also ALLSTAR.

6. *Second ALLFRAME Analysis:* We repeated the same analysis of step 4 but on a catalog in which the stars found in the previous step have been added. The co-adding of the residual images of the second ALLFRAME analysis is shown in Figure 9, in comparison with the co-added image.

7. *CMD selection:* At this point we divided the images by filter, and we ran DAOMATCH separately, referring this time to the files generated by ALLFRAME, and not by ALLSTAR. We ran DAOMASTER requiring that each source must be contained in at least two frames, to reduce the number of

possible spurious detections. At this point we matched the two lists of images using again DAOMATCH and DAOMASTER. This is the standard method to obtain a catalog with instrumental (raw) magnitudes. The obtained catalog contained $\simeq$ 30000 objects, but many of them were spurious sources. The method used to refine the catalog has been quite manual, and implied the use of the TOPCAT java program [8] to perform consecutive selections. The selections were performed analyzing the relations between: (i) the X and Y coordinates to avoid spurious sources at the edges, (ii) the **chi** parameter and the **sharpness** parameter and (iii) the magnitudes and the photometric errors to avoid spurious sources due to the saturation or to computational errors. The chi parameter is defined as chi $\propto \frac{\sigma}{\sigma_{\rm PSF}}$, where σ is the instrumental error and $\sigma_{\rm PSF}$ is the expected error based on the chosen analytic function (see **PSF** task); the sharpness parameter is defined as SHARP $= \sigma^2 - {\sigma_{\rm PSF}}^2$. So, stars tend to have **sharp** values close to 0 and **chi** values close to 1, while galaxies and spurious sources tend to have higher values. Using IRAF as a helping hand we performed all the needed selections and the number of sources reduced from 30000 to $\simeq$ 7000.

8. *Last ALLFRAME Analysis:* The last step of our photometric analysis is the ALLFRAME analysis of the catalog obtained after the selections performed in the seventh step.

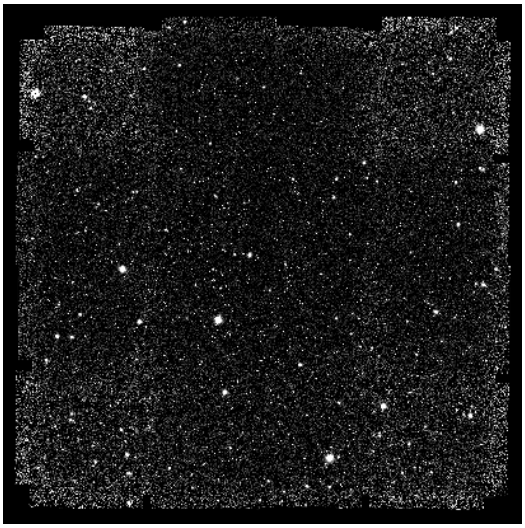


Figure 8: The image obtained co-adding all the 534 images described in Section 2 using MONTAGE2.

[8] http://www.star.bris.ac.uk/ mbt/topcat/

(a)

(b)

Figure 9: *Above*: A zoom on Figure 8; *Below*:A zoom on the image obtained co-adding the residual images from the second ALLFRAME analysis using MONTAGE2. The visible sources are objects not fitted by the code (saturated stars, background galaxies).

After all the steps described above we produced our stellar catalog based on the measures taken with the FourStar Camera. However, this catalog is not calibrated. The first step is to apply the astrometric corrections to the MONTAGE2 image, which all the 534 images refer to. This is possible running the Python codes made available by the Astrometry.net Collaboration, as discussed in Section 3. At this point we needed to choose the calibration catalog. We chose to calibrate in the 2MASS catalog system (Cutri et al. 2003; Skrutskie et al. 2006). However, the 2MASS primary catalog is not deep enough, and so we have to use a deeper local catalog if we want to avoid photometric errors due to saturation and non-linearity. In fact, the majority of the deepest sources of 2MASS match the majority of the brightest ones in our photometry. So, we chose the catalog available from the Peter Stetson's private Database, which has been calibrated with 2MASS. Using DAOMATCH and DAOMASTER, as described above, but with the Stetson's catalog as the master file is possible to calibrate our photometry.

5 Results

Thanks to the 534 images taken with the FourStar Camera presented in Section 2, we derived, as described in Section 3 and Section 4, a star catalog with 7517

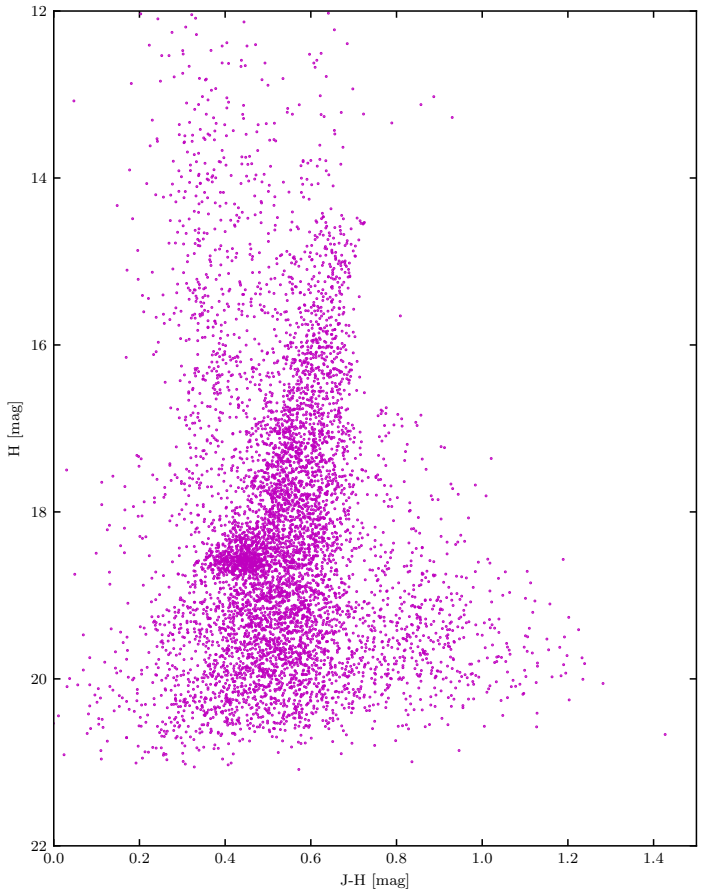


Figure 10: The new H, $J-H$ CMD of Carina dSph, obtained from the catalog which has been produced thanks to the analyses described in Section 3 and 4, for the 534 images taken with the FourStar Camera.

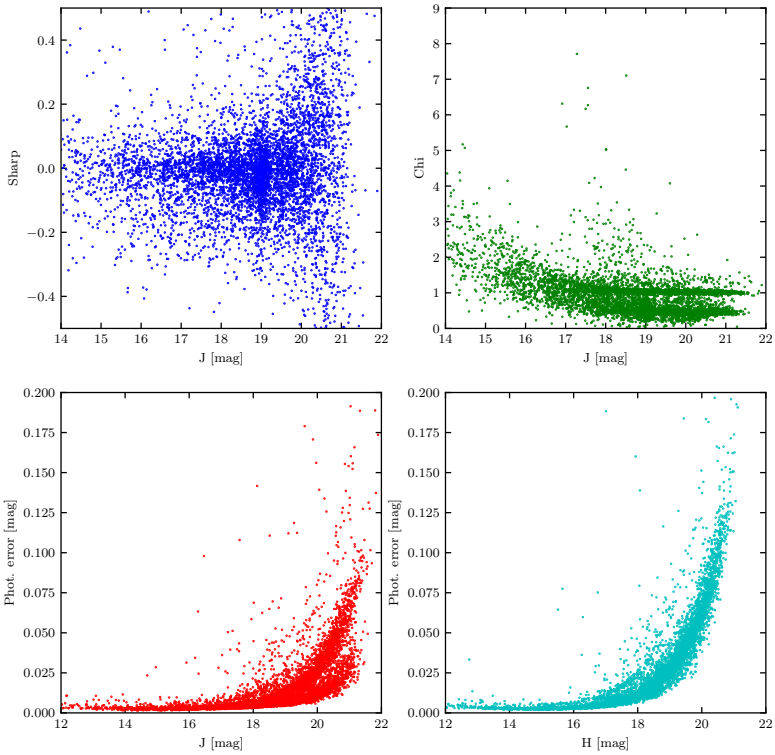
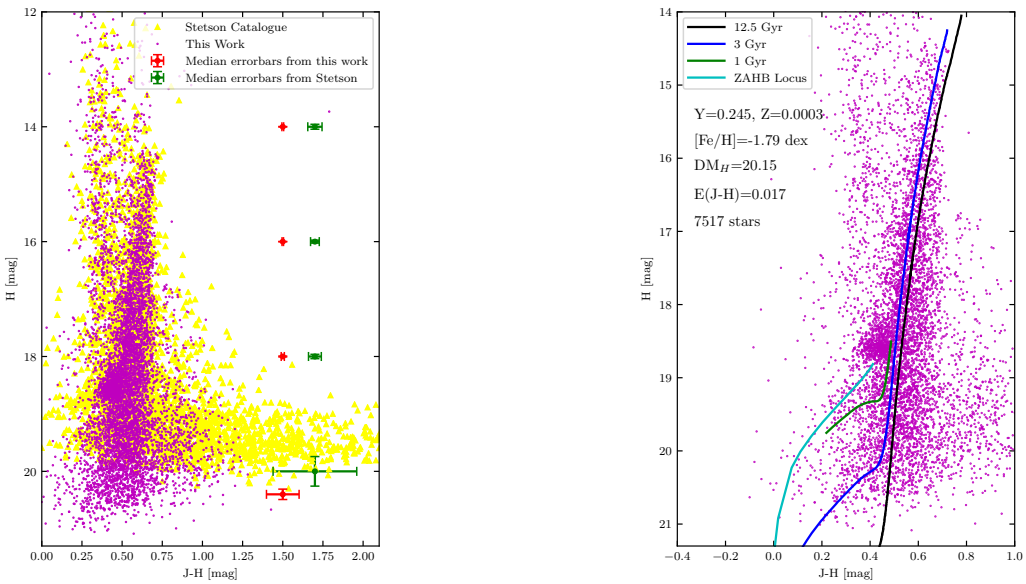


Figure 11: Quality of our photometric reduction. *Above*: Dependence of the parameters **chi** and **sharp** on the J magnitude. *Below*: photometric errors vs J and H magnitudes.

sources. In Figure 10 we present a new NIR H, $J-H$ Color-Magnitude Diagram (CMD) for the Carina dSph. With respect to Figure 2, is possible to notice the RGB, the Red Clump, the Horizontal Branch and even some Sub-Giant stars. The quality of our photometry can be evaluated in Figure 11. Most of our sources have photometric errors < 0.1 mag in the J band, and < 0.15 mag in the H band. The faintest stars in our catalog are in the range of: 21.8 ± 0.1 mag for the J band, and 21.09 ± 0.15 for the H band. This means that our data are ≃ 1.5 mag deeper than the ones from the Peter Stetson's private catalog, and are also more accurate. Moreover, our work is also comparable with the most recent works, as it is possible to observe comparing Figures 10 and 3b. The comparison between our data and the data made available by P.B. Stetson is shown in Figure 12a.



(a) (b)

Figure 12: *Left*: Comparison between the H, $J-H$ CMD obtained from our catalog, in mangenta dots, and the one obtained from the Peter Stetson's private Database, in yellow triangles. Median errorbars (1 σ) are displayed in red, regarding our data, and in green, regarding P.B.S data, at different magnitude levels. It is possible to observe that our data is both deeper and more accurate than the data used for the calibration. *Right*: Our H, $J-H$ CMD is shown in magenta, while the isochrones from the BaSTI Database are shown in black,blue and green. In cyan is showed the ZAHB locus, always from the BaSTI Database.

Moreover, is possible to analyze the new NIR CMD using theoretical isochrones and the ZAHB loci from the BaSTI database, and this kind of analysis has never been proposed in previous literature works on NIR data. Following the same procedure discussed in Stetson et al. 2011, and followed to obtain Figure 2 in Section 1.1, we present the results of our analysis in Figure 12b. The color excess term has been computed thanks to the the IRSA's Dust Reddening and Extinction Calculator, and DM_H=20.15 mag is the apparent distance modulus in the H band (in fact is lower than the one adopted in Section 1.1). The obtained extinction values for $E(B-V) = 0.06$ mag are: $A_J = 0.043$ mag and $A_H = 0.0276$ mag. We found that the experimental data are better represented by canonical, solar-scaled models, and not by alpha-enhanced models. Moreover, the RGB is well fitted by a range of ages from 3 Gyr to 13 Gyr. This is consistent with works which consider mostly an old (>8 Gyr) and an intermediate age (≃3-5 Gyr) populations (Stetson et al. 2011; Bono et al. 2010). Moreover, referring to Figure 12b, it is possible to note a plume of stars below the Red Clump, which are well fitted by 1 Gyr isochrones. This is consistent with works which also consider younger (< 1 Gyr) populations (Monelli et al. 2003; Norris et al. 2017). The isochrones are computed considering a metallicity [Fe/H]=-1.79 dex, and increasing or decreasing the metallicity of $\simeq 0.3$ dex does not produce considerable deviations. This is known as the "Age-Metallicicty degeneracy of the RGB stars" (Monelli et al. 2014). However, the global metallicity used value is consistent with the previous spectroscopic works (Koch et al. 2006) and there is a majority of stars near the Red Clump which are well fitted by a ZAHB model computed using the same metallicity.
Many of these stars may belong to the RR Lyrae instability strip, i.e. He-burning variable stars (period between 0.3 d and 0.8 d) often used as standard candles. To study this possibility, we cross-matched our catalog and the variable stars catalog from Coppola et al. 2015, using the Sky Match algorithm of TOPCat . We found 73 matches, as is possible to see in Figure 13. Among the 73 variables, 67 have measures at least in the J or in the H band and are represented in the Bailey diagram (V band pulsation amplitude vs pulsation period) in Figure 14.

Among these 67 stars, 33 are RRab variables (fundamental harmonic and sawtooth-like light curve) and 7 are RRc variables (first overtone and sinusoidal-like light curve). As we saw RRL could be used as standard candles, so for this kind of variables could be interesting to compute mean magnitude values and verify the so-called period-luminosity-metallicity relation (PLZ). To understand what is a PLZ we need to consider both theoretical and observational aspects. From the theoretical point of view, we need to remember that RRL are low-mass, old (> 10 Gyr) He-burning stars and so their luminosity and effective temperature is highly dependent on the He core mass value which typically is $\simeq 0.5 M_{\odot}$. Radial velocities studies showed that RRL are subjected to radial variations which in turn

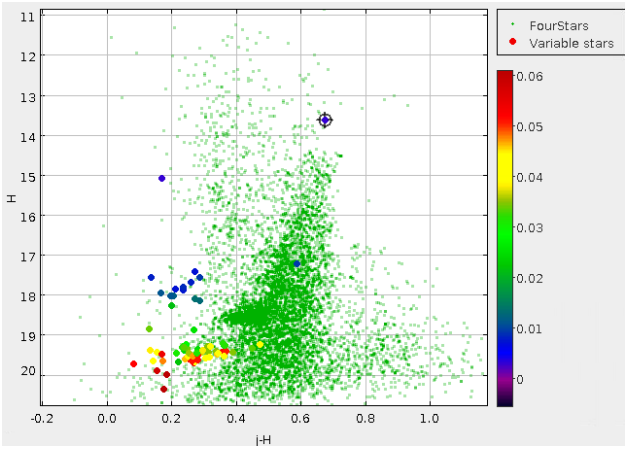


Figure 13: The FourStar NIR CMD is shown with green dots, while the matched variable from Coppola et al. 2015 are shown with a rainbow color map, whose intensity is proportional to the photometric error.

could cause a luminosity variation according to $L = 4\pi\sigma R^2 T^4$. However, a star normally is in stable hydrostatic equilibrium, i.e. if its outer layers expand the density and temperature decrease, leading to a pressure and an opacity decrease which in turn leads to a compression thanks to the action of gravity. These are the radial variations or pulsations of the stars, which however, without an energy source are destined to be damped (Karttunen et al. 2017). The radiative flux from the interior could be a source, if and only if it were preferentially absorbed in particular regions, which have been identified with the partial ionization zones of H and HeII. In these zones the opacity becomes larger when the gas is compressed, making the star unstable against pulsations. This is called κ-mechanism and has been introduced by Sir Arthur Eddington in 1926 (Eddington 1926). That is why these kind of stars need to be in a precise place in the CMD called instabilty strip, where the ionization of H and He is more efficient. The κ-mechanism is not the only mechanism which could drive pulsation. In fact, the presence of a ionization layer could modify the adiabatic coefficient causing the outward flux to decrease and the layer itself to retain the energy it receives from below. This is called the γ-mechanism (King et al. 1968) and an accurate theoretical description of RRL pulsations must take in account both the cited mechanisms and the relation between them.

From the observational point of view, RRab follow the relation $P \propto \frac{1}{\sqrt{\rho}}$ (Ritter 1879), which express that for a radial pulsator the period P is inversely proportional to the bulk denisty ρ. So, using also the luminosity relation we could write:

$$\log P_{\mathrm{RRab}} = const. + \frac{3}{4}\log L - \frac{1}{2}\log M - 3\log T \tag{1}$$

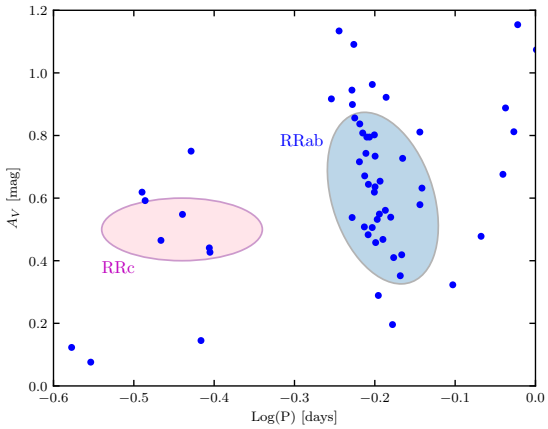


Figure 14: The 67 variable stars which are common to our catalog and to the one produced by Coppola et al. 2015 are represented with blue dots in the so-called Bailey diagram, i.e. V band amplitude versus pulsation period. With blue and pink ellipses is represented the parameter space usually occupied by RRab and RRc variables.

This is essentially the core of a period-luminosity relation, as the one first discovered by Henrietta Swan Leavitt observing the Cepheids in the Large Magellanic Cloud. However, it is only an approximation and an accurate expression needs to take in account internal and evolutionary models and also the role of metallicity, becoming a PLZ (Marconi et al. 2015). Using theoretical models a relation could be derived also for RRc. The only observational quantity is the period so we need to connect L and T to the magnitudes and to the color indices using the bolometric correction, while M is generally estimated from evolutionary models (Marconi et al. 2003). In the NIR bands the bolometric correction will give a linear relation between magnitudes and temperatures (in contrast to the V band), and so using Eq. 1 is possible to observe that we will deal with a linear relation between period and near infrared magnitudes.
There is a single work in literature where Carina dSph RRL NIR mean magnitudes have been computed (Karczmarek et al. 2015), even if there are many reasons to study RRL PLZ in the NIR domain, as:

- NIR observations are less affected by the reddening
- The effect of metallicity on the luminosity is quite small in the IR domain (Sollima et al. 2008; Marconi et al. 2015)
- New theoretical predictions could be verified, as nonlinear convective pulsation models (Bono et al. 2003)

To pursue this goal, first we need to compute the experimental phase points. Using IRAF we converted the UT times of the exposures, expressed in the header of the 534 images in Heliocentric Julian Date (HJD). At this point we used the information collected in the .cor and .mag files produced by DAOMATCH and DAOMASTER, to connect every photometric value to the right exposure time in HJD. Having only a night of observations, for most of the matched variables we have $\simeq$30 measures in J, and $\simeq$ 20 in H. Due to the observational strategy the J band measures are more numerous and so span a larger time window than the H band measures, as explained in Section 2. Having few data blocks is impossible to recover a full RR Lyrae light curve, so is reasonable to bin the data and find a way to obtain an estimate for the mean magnitudes. To bin the data we used an IDL program which performs a weighted sigma-clipped (the data are iteratively kept or rejected in terms of standard deviation) mean having as a prior the maximum time interval to be considered as a bin. We chose 0.005 days because this time interval contains all the group of measures and is smaller than the time interval between two distinct group of measures. Moreover, we excluded measures with photometric errors greater than 0.2 mag to improve the quality of the mean magnitude values.

After the analysis described above, we found that the majority of the J measures correspond to two weighted phase points in most of the cases. For two variables, labelled as 160980 (V58) and 5001226 (V183) the measures resulted in three phase points. However, most of the H measures resulted in one single phase point, and many others didn't have any point. A series of experimental and binned light curves in the J band are represented in Figures 15 and 16.

The binning process is essential to compute the mean magnitudes. There are different ways to obtain mean magnitudes and one of the most accurate, but also most computational expensive, is the template analysis (Braga et al. 2019). However, to obtain a PL relation it could be used another simpler method where the mean magnitudes are intensity averaged.
The method requires three steps:

1. Convert binned magnitudes in intensity using the famous Pogson relation with an arbitrary zero-point ($c_0 = 0$), $L = L_0 \cdot 10^{\frac{m-c_0}{2.5}}$

2. Average the intensities using a weighted mean

3. Convert the average value in magnitude using again the Pogson relation, $m = 2.5 \cdot \log \frac{\overline{L}}{L} + c_0$

The period and mean magnitudes of the 67 variables are listed in Table 4. To verify that the averaging process didn't produce corrupted measures we plot the

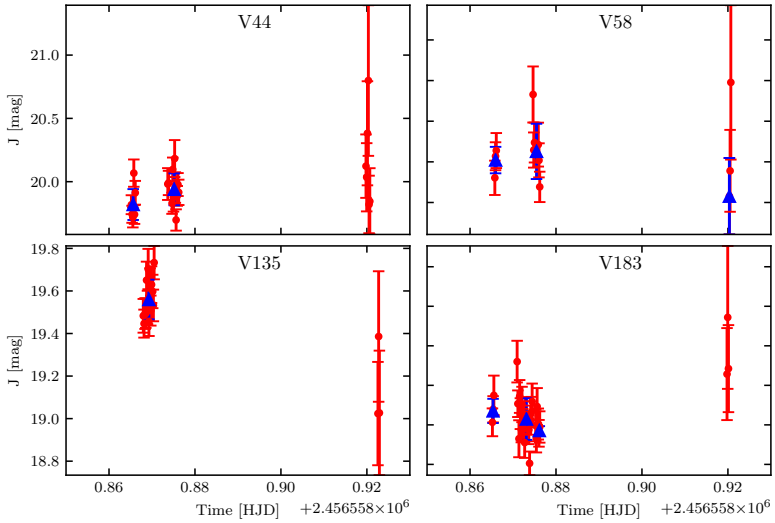


Figure 15: Light curves, J band magnitudes vs HJD, derived from our photometric reduction (in red dots) and binned light curves (in blue triangles) for four selected variables: V44, V58, V135, V183 (Coppola et al. 2015). Is possible to observe the criteria of the binning process as measures with large errors are not considered.

mean magnitude in the H, $J-H$ CMD in Figure 17, and we observe that the result is very similar to Figure 13.

We represented the values listed in Table 4 in Figure 18, in a mean magnitude vs period diagram, considering only the RRab and RRc variables. RRab and RRc follow different PLZ relations due to the different pulsation properties (fundamental and first overtone pulsations). However, is possible to obtain a single relation both for the J and H bands considering the relation between the median periods of RRab and RRc. Different literature works converged on the relation $\langle P_{\rm RRab}\rangle \simeq \frac{3}{4}\langle P_{\rm RRc}\rangle$, and so in samples with a small number of RRc it is customary (Coppola et al. 2015) to fundamentalize the RRc stars using the relation:

$$\log P_{\rm fund} = \log P_{\rm RRc} + 0.127 \tag{2}$$

where $P_{\rm fund}$ is the fundamentalized period and $P_{\rm RRc}$ is the first overtone pulsation period of the RRc variables. It is possible to notice that the data follow a clear linear relation in the J band, while the same is not as clear in the H band. This is caused both by the observational strategy, explained in Section 2, and both by the worse photometric quality in the H band with respect to to the J band, as is possible to observe in Figure 11. Using the LabPlot software[9] fitting algorithm, we fit the data using a linear model, with an intercept A and a slope B, and we

[9]https://labplot.kde.org/

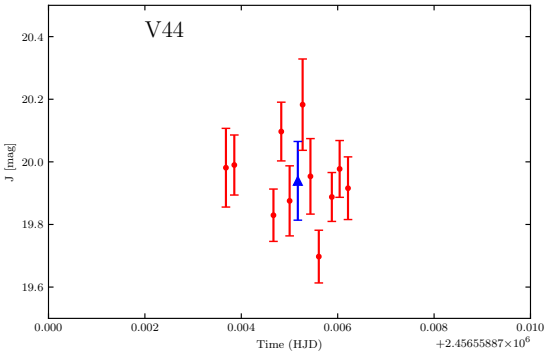


Figure 16: A zoom on the left upper pannel of Figure 15, which could be useful to understand the chosen binning strategy.

compared them with the theoretical PLZ relations from Marconi et al. 2015, which in the i-band could be expressed as:

$$\langle M_i \rangle = a + b \log P + c[\mathrm{Fe/H}] + \mathrm{DM}_i \tag{3}$$

where $\langle M_i \rangle$ is the intensity averaged mean magnitude in the i-band, P is the pulsation's period and [Fe/H] is the metallicity. We adopted [Fe/H]=-1.79 dex, to be consistent with our previous analysis. The results of the fit and the used theoretical parameters are listed in Table 3.

	A [mag]	B [mag]	a [mag]	b [mag]	c [mag]
J band	19.30 ± 0.04	- 1.5 ± 0.2	-0.50 ± 0.01	-1.90 ± 0.03	0.18 ± 0.01
H band	19.1 ± 0.1	- 1.4 ± 0.6	-0.76 ± 0.01	-2.22 ± 0.02	0.18 ± 0.01

Table 3: Parameters of the best-fitting linear models, labelled with A and B, and the parameters of the theoretical PLZ relations from Marconi et al. 2015.

It is possible to observe that the best-fitting model is close to the theoretical PLZ relations, both in the J and in the H band. In fact, from the results listed in Table 3, is possible to notice that the slopes are of the same magnitude even if they are not consistent. However, the agreement is considerable better in the J band than in the H band, where only three RRc have a phase point, and so an important region of the $\langle H \rangle$-log P diagram is less populated. This is due mostly to the observational strategy, in fact only a small region of the Carina dSph has been observed twice in a time interval of few hours, and it has been observed only in the J band; moreover, that is why we found only 73 variables out of 117, during the cross-match with the list from Coppola et al. 2015. With this caveat in mind,

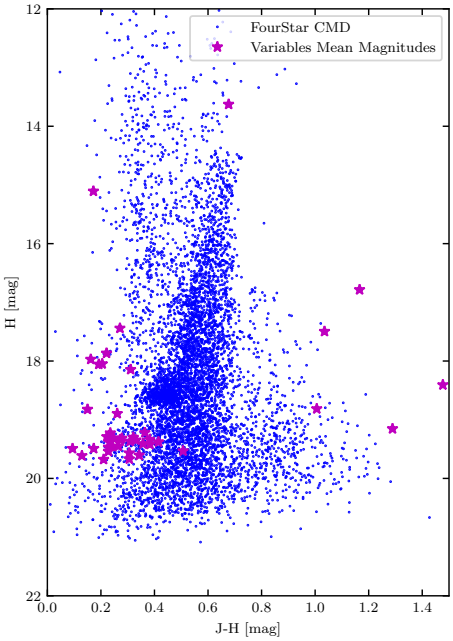


Figure 17: The 67 variables mean magnitudes values are represented with magenta stars in the H, $J-H$ CMD. For a comparison we represent with blue dots the data from Figure 10.

and knowing that J photometry is more accurate we could focus on the J data and use together the data with the theoretical curves to obtain an estimate of the distance. The best-fitting linear model gave an intercept B = 19.30 ± 0.04 mag, and the theoretical values are $a = -0.5$ mag and c[Fe/H] = −0.32 mag, giving a value for the apparent distance modulus of $\mathrm{DM}_J = 20.12 \pm 0.04$ mag. The extinction in the J band is $A_J = 0.043$ mag, and so a simple estimate of the true distance modulus is $\mathrm{DM}_0 = 20.08 \pm 0.04$ mag, which is consistent both with the value from Dall'Ora et al. 2003, and both with the most recent works of Coppola et al. 2015 and Karczmarek et al. 2015, and consistent with the value used in the isochrones analysis.

6 Summary and Conclusions

We presented new and unpublished NIR data of Carina dSph. The dataset consisted of 534 images taken thanks to the FourStar NIR camera installed at the

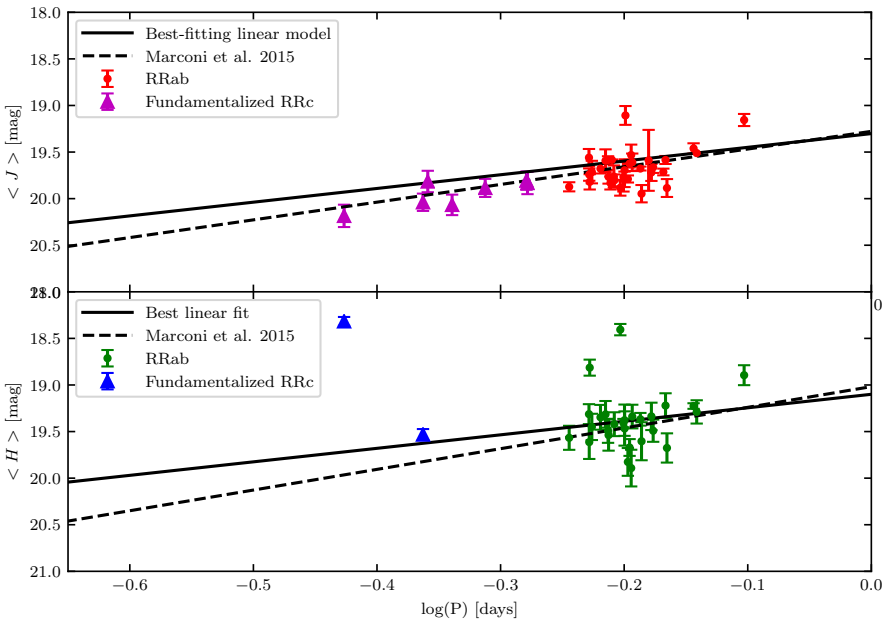


Figure 18: Empirical and theoretical P-$\langle J \rangle$ (above) and P-$\langle H \rangle$ (below) relations of the Carina dSph RRLs. RRab are represented with dots, while fundamentalized RRc are represented with triangles. Best-fitting linear models are represented with continuous black lines while the theoretical relations from Marconi et al. 2015 are represented as dashed black lines.

6.5-m Baade Telescope (Las Campanas, Chile), 240 in the H band and 294 in the J band. We discussed in the detail the pre-reduction strategy, performed thanks to the IRAF software, and the reduction strategy performed thanks to the DAOPHOT/ALLSTAR/DAOMATCH/DAOMASTER/ALLFRAME suites. The result of our work is comparable in depth and quality to other NIR surveys, and is $\simeq$ 1.5 mag deeper than the catalog used for the calibration. The obtained catalog contained 7517 stars, with photometric errors under the 0.1 mag for the majority of stars in the J band, and under the 0.15 mag in the H band. We obtained and discussed the H, $J-H$ CMD and we compared it with the isochrone curves from the BaSTI database. The results are consistent with the precedent litearture works which are based on optical photometry. We found that the data is well fitted by an intermediate age population of 3 Gyr and an old population of 12.5 Gyr. The RGB is well fitted in this range of ages, and is weekly dependent on metallicity spreads. We considered a metallicity value of [Fe/H]=-1.79 dex, which was used also to compute the ZAHB which fit well the data. Moreover, we found evidence for a young population of 1 Gyr.
Then, our analysis focused on the variables stars. We cross-matched our catalog with the list of variables from Coppola et al. 2015, and we found 73 matches. We obtained the binned light curves and the intensity averaged mean magnitudes for every variable which had at least a measure in both the J and H bands. We studied the NIR PLZ relations and compared them with the theoretical relations from Marconi et al. 2015. At the end we found a true distance modulus of Carina dSph of $DM_0 = 20.08 \pm 0.04$ mag, which is consistent with previous and most recent literature works.

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ID	Type	P [days]	A_V	$\langle J \rangle$ [mag]	$\langle H \rangle$ [mag]
V 25	Bl	0.596	0.856	19.621 ± 0.004	19.388 ± 0.182
V 27	AC	1.02	0.971	18.531 ± 0.061	17.496 ± 0.076
V 29	AC	0.718	0.811	18.247 ± 0.077	18.055 ± 0.054
V 31	Bl	0.646	0.468	19.54 ± 0.055	19.308 ± 0.139
V 32	ab	0.615	0.743	19.832 ± 0.07	-99.0 ± -99.0
V 40	c	0.393	0.441	19.807 ± 0.098	-99.0 ± -99.0
V 41	AC	1.038	1.215	18.087 ± 0.021	17.866 ± 0.034
V 44	ab	0.626	0.963	19.883 ± 0.085	18.406 ± 0.061
V 47	c	0.324	0.619	20.039 ± 0.093	19.532 ± 0.059
V 49	ab	0.682	0.419	19.585 ± 0.043	19.221 ± 0.132
V 57	ab	0.613	0.671	19.766 ± 0.073	19.538 ± 0.167
V 58	ab	0.619	0.644	19.804 ± 0.064	19.421 ± 0.128
V 60	ab	0.609	0.808	19.593 ± 0.122	19.315 ± 0.143
V 61	Bl	0.621	0.795	19.637 ± 0.114	-99.0 ± -99.0
V 65	ab	0.652	0.922	19.947 ± 0.093	19.604 ± 0.204
V 67	ab	0.604	0.716	19.678 ± 0.052	19.347 ± 0.133
V 68	ab	0.679	0.352	19.714 ± 0.036	16.642 ± 0.023
V 73	ab	0.57	1.134	19.873 ± 0.049	19.567 ± 0.129
V 75	ab	0.591	0.538	19.742 ± 0.066	19.613 ± 0.181
V 77	Bl	0.604	0.837	19.7 ± 0.059	19.457 ± 0.139
V 84	ab	0.617	0.795	19.584 ± 0.005	16.536 ± 0.015
V 85	ab	0.641	0.654	19.61 ± 0.094	19.339 ± 0.127
V 87	AC	0.856	0.478	18.1 ± 0.04	-99.0 ± -99.0
V 90	ab	0.631	0.636	19.793 ± 0.083	19.378 ± 0.168
V 91	ab	0.718	0.579	19.461 ± 0.055	19.226 ± 0.03
V 92	ab	0.63	0.619	19.766 ± 0.158	19.409 ± 0.045
V 115	AC	1.011	0.691	17.709 ± 0.026	19.597 ± 0.18
V 116	ab	0.683	0.727	19.886 ± 0.096	19.676 ± 0.157
V 122	ab	0.631	0.734	19.695 ± 0.117	19.467 ± 0.184
V 124	ab	0.592	0.899	19.82 ± 0.082	18.814 ± 0.086
V 125	ab	0.594	1.091	19.715 ± 0.121	19.45 ± 0.141
V 126	Bl	0.557	0.917	19.975 ± 0.139	19.669 ± 0.129
V 127	Bl	0.626	0.506	19.621 ± 0.094	19.3 ± 0.181
V 129	AC	0.63	0.802	18.53 ± 0.077	-99.0 ± -99.0
V 135	ab	0.591	0.945	19.56 ± 0.093	19.313 ± 0.107
V 138	ab	0.639	0.549	19.532 ± 0.112	19.892 ± 0.197
V 141	ab	0.635	0.532	19.789 ± 0.038	19.826 ± 0.149
V 142	c	0.364	0.548	19.885 ± 0.097	-99.0 ± -99.0
V 144	c	0.393	0.427	19.832 ± 0.12	-99.0 ± -99.0
V 148	c	0.327	0.592	19.816 ± 0.115	17.795 ± 0.043
V 149	AC	0.918	0.888	19.206 ± 0.042	17.325 ± 0.028
V 151	c	0.342	0.465	20.067 ± 0.11	-99.0 ± -99.0
V 153	ab	0.66	0.539	19.589 ± 0.327	14.063 ± 0.009
V 158	ab	0.632	0.458	19.107 ± 0.101	17.208 ± 0.038
V 178	AC	1.016	1.244	18.133 ± 0.022	17.971 ± 0.038
V 179	ab	0.664	0.196	19.719 ± 0.091	19.337 ± 0.147
V 181	c	0.279	0.076	20.185 ± 0.12	18.317 ± 0.047
V 182	ab	0.789	0.323	19.156 ± 0.065	18.895 ± 0.107
V 183	ab	0.612	0.508	19.586 ± 0.041	19.492 ± 0.129
V 187	AC	0.95	1.154	18.368 ± 0.061	-99.0 ± -99.0
V 190	AC	1.165	1.294	18.067 ± 0.004	19.645 ± 0.055
V 191	ab	0.65	0.561	19.677 ± 0.021	19.368 ± 0.068
V 194	FV	0.265	0.123	15.278 ± 0.007	15.106 ± 0.014
V 195	ab	0.619	0.483	19.763 ± 0.119	17.067 ± 0.047
V 196	ab	0.666	0.41	19.664 ± 0.06	19.491 ± 0.119
V 200	ab	0.637	0.289	19.624 ± 0.042	19.671 ± 0.09
V 201	ab	0.722	0.632	19.511 ± 0.004	19.289 ± 0.125
V 203	AC	0.94	0.812	18.253 ± 0.031	18.048 ± 0.035
V 205	AC	0.383	0.145	18.973 ± 0.06	18.823 ± 0.126
V 216	AC	1.079	0.303	17.951 ± 0.061	16.785 ± 0.016
V 217	AC	0.911	0.676	17.966 ± 0.033	18.457 ± 0.102
V 219	AC	1.366	0.387	17.712 ± 0.001	17.441 ± 0.03
V 220	EB	0.373	0.75	13.304 ± 0.083	18.393 ± 0.071
V 222	LP	329.0	99.0	14.303 ± 0.017	13.626 ± 0.022
V 230	AC	1.003	1.074	18.456 ± 0.042	18.145 ± 0.045
V 231	EB	2.383	99.0	20.443 ± 0.154	19.154 ± 0.097
V 234	EB	0.527	99.0	17.832 ± 0.045	19.28 ± 0.093

Table 4: Periods, Visual Amplitude and new mean magnitudes of the 67 variables found from the cross-match between our catalog and the one from Coppola et al. 2015. In order: "Bl" refers to RRL Blazhko Variables, "AC" refers to Anomalous Cepheids, "FV" to a generic field variable, "LP" to a long period variable and "EB" to eclypsing binaries.